

Advanced Views

Q1. v_ManagerDepth

Create a view that shows each doctor's full name, specialization name, and the number of supervisors above them in the hierarchy (using **superssn** chain).

Q2. v_CrossSpecDoctors

Create a view listing doctors who have performed surgeries outside their specialization. Show doctor name, their specialization, and count of such surgeries.

Q3. v_InvalidSurgeryLocations

Create a view showing surgery name, its location, specialization name, and a status column ("VALID" / "INVALID") depending on whether the location exists in **spec_locations**.

Q4. v_OverloadedManagers

Create a view showing each manager's name, number of doctors they supervise, and total surgery hours performed by their team.

Q5. v_DependencyBurden

Create a view showing doctor name, salary, number of dependents, ordered by a burden index (dependents ÷ salary).

Advanced Stored Procedures

Q6. sp_ReassignToBalancedSpec

Create a procedure that moves a doctor to the specialization with the fewest doctors and updates their supervisor to that specialization's manager.

Q7. sp_RedistributeSurgeryHours

Create a procedure that, if a doctor has more than 30 hours on a surgery, reduces their hours by 20% and distributes the reduced time equally among other doctors working on the same surgery.

Q8. **sp_FixSupervisorCycles**

Create a procedure that detects cycles in the supervisor chain (**superssn**) and breaks them by setting one supervisor reference to NULL.

Q9. **sp_ReplaceManager**

Create a procedure that replaces the manager of a specialization. Ensure the new manager belongs to the same specialization and update subordinates accordingly.

Q10. **sp_DeleteDoctorCascade**

Create a procedure to safely delete a doctor by:

- Reassigning dependents to their supervisor
 - Reassigning their surgeries to another doctor in the same specialization
 - Replacing them if they are a manager
 - Then deleting the doctor
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Combined View & Procedure (Practice)

Q11. **v_DoctorRanking + sp_RewardTopDoctors**

Create a view ranking doctors within each specialization based on total surgery hours. Then create a procedure that gives a 15% raise to the top 2 doctors in each specialization.

Q12. **v_LocationViolations + sp_FixLocations**

Create a view showing surgeries with invalid locations. Then create a procedure that updates such surgeries to a valid location from **spec_locations**.

Q13. **v_WorkloadStats + sp_NormalizeWorkload**

Create a view showing average workload per specialization and each doctor's workload. Then create a procedure that adjusts doctor hours if they are too high or too low compared to the average. Make a suitable threshold in either direction.

Q14. sp_RecommendHiring

Create a procedure that lists specializations needing hiring based on:

- Fewer than 2 doctors OR
 - Average workload greater than 25 hours
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Q15. sp_EnforceSalaryRules

Create a procedure that ensures every manager earns at least as much as their subordinates. If not, update the manager's salary.